How the 32X Games Use the Processors

Oct 13, 2026 · @erexx

The shared model (OpenLara, Jazz, Hurrican and most of the CD games)

In the CD games the master SH2 runs the game, the Sega CD does disc and sound, and the 68000 only copies data and reads pads.

Processor	Its job
Master SH2	Runs the whole game: logic, drawing, timing from the vblank interrupt. Sends every request to the Sega CD by writing a command to COMM0 and waiting for COMM0 = 0.
Slave SH2	Optional helper. It polls for a job, never takes interrupts, and signals done through COMM or shared SDRAM.
Genesis 68000	Chilly Willy's command listener (hw_md.s). Reads the pads in its vblank, copies data into the RAM cart with RV = 1, turns the Genesis text/display on or off. It plays no sound and draws nothing in game.
Sega CD Sub-CPU	The service loop: CD file loading, CD music, PCM sound effects, backup RAM.

Identical in OpenLara, Jazz and Hurrican:

- **Command numbers:** 38 open file, 46 chunk to cart, 48 pad + sound effect, 45/54 music on/off, 52 Genesis display, 53 load PCM effects, 61 save to cart. Status goes in COMM12 (0x4B10 = game running).
- **Loader and support code:** the same SH2 loader (Jazz and Hurrican use OpenLara's), and the same hw_md.s and Sub-CPU code (only comments differ).
- **RAM cart:** the program sits at 0, data packs above it, saves at 0x3F0000. The SH2 never writes the cart; the 68000 does every write with RV.
- **Sound:** CD audio music and Sega CD PCM effects. None of the three uses PWM (except OpenLara's movies) or the YM2612.
- **Graphics:** 32X 256-colour mode, double-buffered. The Genesis VDP shows nothing in game.

Where it all came from: Kobo (kobo32x_hw)

Kobo Deluxe is the root of this stack. Its boot sequence was reverse-engineered from the *Night Trap* CD32X disc (the two PDFs in the repo), and the Sub-CPU command set (38, 46, 48, 45/54, 52, 53, 56–59) starts here. kobo32x_hw is the real-hardware build of kobo32x: same SH2 code, identical hw_md.s, but it requires the cart (KOBO_NOCART 0); build.sh refuses to make a hardware disc in no-cart mode.

Processor	In Kobo (kobo32x_hw)
Master SH2	The whole game. Runs from SDRAM. 32K-colour for the splash, 256-colour in game.
Slave SH2	Unused (while(1)).
Genesis 68000	The hw_md.s listener: pads, Genesis text, RV copies into the cart.
Sub-CPU	CD file loading, CD music (3 tracks), PCM effects, backup RAM.

Everything later added on top of this base: OpenLara's level slot and fast start, Jazz's job mechanism, Hurrican's slave logic.

Where they differ: the slave SH2

OpenLara splits the drawing, Hurrican splits logic from drawing, and Jazz and Kobo stay single-CPU.

Game	Slave SH2 does
OpenLara	Draws its share of the 3D frame (split screen) and puts the finished frame on screen. During movies it plays the sound to PWM. Gets jobs through COMM4.
Hurrican	Runs the game logic while the master draws tiles. Gets jobs through a shared SDRAM job struct, polled by both. Switchable in the pause menu.
Jazz	Nothing. The same job mechanism as Hurrican is built in but switched off (OJ32X_SLAVE_DRAWS = 0).
Kobo	Nothing.

How they use the 4 MB RAM cart

Every CD game fills the cart the same way: the Sega CD reads the file, the 68000 copies it in 32 KB at a time with RV = 1, and the SH2s then read straight from the cart like a cartridge. Nothing is copied in while a level plays. The cart keeps its contents without power, so most games check what's already there and skip the copy.

Game	0 – 1 MB	Middle	Upper	Last 64 KB (0x3F0000)
OpenLara	Program (0–0.5 MB)	Caves / title (0.5–2 MB), kept across power-off	Current level or movie (2–3.94 MB)	Settings + saved game
Jazz	Program	Common data (1–2.5 MB), kept across power-off	Current planet (2.5–3.94 MB)	Saved game
Hurrican	Program	Resident pack (1–2.5 MB), kept across power-off	Current stage (2.5–3.94 MB), not reloaded if already there	Saves + settings
SuperTux	Program	Shared data + title level (1–1.6 MB)	Levels (1.6–3.94 MB), each in its own place, kept until room is needed	Platform save area (game saves in Sega CD backup RAM)
Heretic	Engine + tables (1.45 MB pack)	Level region from about 1.5 MB, loaded per level	–	No saves
Devilution	Game program	Level/hero slots only planned (town only so far)	–	Saves in Sega CD backup RAM
OpenTyrian (CD boot disc)	Whole cartridge image, copied every boot	–	–	Kept free (saving fails on this disc)
Kobo (kobo32x_hw)	– (program runs from SDRAM)	Graphics pack KOBOGFX.BIN at 2 MB (slot 16 × 128 KB, up to 1 MB), copied at boot	–	Saves in Sega CD backup RAM

OpenLara, Jazz and Hurrican share the same three-part split:

1. The program at 0.
2. A "resident" block that's checked at start and only reloaded when it's missing (A+B+C forces a reload in OpenLara and Jazz).
3. One slot for the current level.

Saves go in the last 64 KB, written by the 68000 in 4 KB pieces. Kobo used the cart only for its graphics. Its "file number × 128 KB" slot rule is still what the Sub-CPU's chunk command uses in every game.

No RAM cart: Rev, Another World, Blake Stone, OpenTyrian's main release and Sonic 32X Plus run from a normal ROM cartridge. kobo32x (the Fusion build) reads its graphics straight from the CD. Blake Stone and Rev save to cartridge SRAM; Tyrian saves to Sega CD backup RAM.

The other games, briefly

Game	Stack	Slave SH2
SuperTux, Devilution, Heretic, Rev	Same CD stack and command set (from Kobo)	SuperTux: Jazz-style job mechanism, off. Heretic: renders from a queue. The rest: idle.
Tyrian, Another World	Cartridge (cart_md.s answers the same command numbers)	Mixes PWM sound
Blake Stone	D32XR stack	Draws part of the view and mixes PWM; the 68000 plays YM2612 music
Sonic 32X Plus	Its own (drx's Sonic 32X); the opposite design, see below	PWM drum samples

Sonic 32X Plus: the opposite design

Here the 68000 runs the game and both SH2s take orders

Processor	Its job
Genesis 68000	Runs all of Sonic 1 plus Tails' logic. Draws the Genesis planes and sprites. Plays the music on YM2612/PSG through the Z80 sound driver. Sends 4 COMM words to the master SH2 every frame: frame toggle, camera, Tails' animation frame, Tails' position.

Processor	Its job
Master SH2	Draws Tails, the sky gradients and the credits plasma in 256-colour, layered over the Genesis picture. C code on YATSSD. Polls COMM0–6, syncs on vblank.
Slave SH2	Sound only. Plays Sonic 1's drum samples through PWM, reading play/stop/volume from COMM8/10 inside its PWM interrupt.
Sega CD (optional)	CD audio on the credits screen only. Single-letter commands (P play, S stop, Z pause…) through the Sega CD's own communication port.

What it shares with the other games: COMM registers used as one-way mailboxes, a toggle bit as the frame heartbeat, a handshake bit for acknowledgement, and no command interrupts.

Bottom line

- The CD games run on one design, started by Kobo: the master SH2 does the game, the Sub-CPU does CD and sound, and the 68000 copies data and reads pads.
- The RAM cart is used like a cartridge that the CD fills: program, a resident block kept across power-off, one level slot, saves at the top.
- Sonic 32X Plus is the only inverted one: the 68000 is the game and the SH2s are its graphics and sound helpers.
- The only real variable among the rest is the slave SH2. It splits the drawing (OpenLara), runs the logic (Hurrican), mixes PWM (cartridge games), or sits idle (Jazz, Kobo).
- The cheapest next win would be switching on the slave job mechanism that Jazz and SuperTux already have built in.